

Article

Stability Analysis in a Mathematical Model for Allergic Reactions

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Abstract: We present a mathematical model that captures the dynamics of the immune system during allergic reactions. Using delay differential equations, we depict the evolution of T cells, APCs, and IL_6 , considering cell migration between various body compartments. The biological discussions and interpretations within the article revolve around drug desensitization, highlighting one potential application of the model. We conduct stability analysis on certain equilibrium points, demonstrating stability in some cases and only partial stability in others. Numerical simulations validate the theoretical findings.

Keywords: mathematical model; immune system dynamics; allergic reactions; delay differential equations; T cells; APCs; IL; cell migration; drug desensitization; stability analysis; equilibrium points; numerical simulations; partial stability analysis

MSC: 92D25; 34K20; 92C60

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1. Introduction

An allergic response stems from an exaggerated reaction of the immune system. When the body encounters an allergen, it triggers the production of a substantial amount of IgE (immunoglobulin E) antibodies, responsible for instigating allergic reactions. Allergy symptoms can vary widely, from itching, rash, swelling, and cramps to the severe and potentially life-threatening anaphylactic shock.

Among the myriad potential allergens, drug-induced allergic reactions pose significant challenges. As remarked in [1], the widespread use of chemotherapy drugs has escalated hypersensitivity reactions (HSRs) globally. These specific allergic responses can endanger lives, restricting the use of primary medications and jeopardizing both patient survival and quality of life. Re-exposure to allergic-induced medications has tragically resulted in fatalities. A study of such drug-induced allergies was carried out in [2]. In this paper, we focus on modeling general allergic reactions.

The immune system comprises a complex network of biological elements and processes crucial for defending the body against diseases. Central to this system are white blood cells (lymphocytes), categorized into myeloid cells and lymphoid cells. Myeloid cells encompass various types, such as neutrophils, basophils, dendritic cells, and macrophages. Lymphoid cells consist of T cells, B cells, and natural killer cells.

T cells can be further categorized into helper T cells, memory T cells, cytotoxic T cells, and regulatory T cells. Regulatory T cells (Treg) play a pivotal role in modulating immune system activation and preventing autoimmune responses.

Antigen-presenting cells (APCs) aid in activating T cells. They are classified as professional APCs (including dendritic cells, macrophages, and B cells) or non-professional APCs (found in various nucleated cell types within the body). Upon contact with an allergen, APCs capture the allergen and present it to naive T helper cells.

The immune system relies significantly on cytokines, small proteins that coordinate immune responses. Interleukins, a type of cytokine, facilitate communication among white blood cells. These cytokines, produced predominantly by white blood cells, regulate the differentiation of other cells and can either stimulate or inhibit their functions. Notably, the same cytokine may be produced by various cell types.

Immunoglobulins, or antibodies, are generated by plasma cells derived from mature B cells. There are five primary antibody classes—IgA, IgD, IgE, IgG, and IgM—each with distinct structures, functions, and responses to antigens. Elevated levels of IgE are associated with allergic reactions.

Maintaining immunological balance is crucial in managing allergies, which involves a well-balanced Type 1 T helper (Th1) and Type 2 T helper (Th2) response. T helper cells, major cytokine producers, influence this balance. Certain Th2-type cytokines correlate with IgE production, suggesting that an imbalance toward Th2 may lead to allergies [3].

Mature T cells exhibit autocrine action, regulating their population and suppressing other cells via their cytokines. Th1 and Th2 cells suppress each other, while Treg cells suppress both Th1 and Th2 cells without being suppressed themselves.

Interleukin-6 (IL-6) is a cytokine known for promoting Th2 cell differentiation and inhibiting Th1 and Treg growth and differentiation in response to allergens [4]. It can be produced by both professional antigen-presenting cells (APCs) and T helper cells [4,5]. The authors of [6] were the first to demonstrate the influence of IL-6 on tilting the Th1/Th2 balance toward Th2.

Drug desensitization, a process for inducing temporary immunological tolerance in individuals hypersensitive to certain drugs, has proven safe and effective in improving clinical outcomes. This method involves gradually administering increasing doses of the drug [7,8]. Rapid IgE desensitization inhibits IL-6 [8], and studies [9] showed a 75% reduction in IL-6 production post-desensitization.

According to [10], the body consists of two compartments: the central compartment comprising the blood and well-perfused organs, and the peripheral compartment encompassing poorly perfused tissues and organs.

As argued in the paper by Fishman and Segel [11], effective immunotherapy for allergies necessitates considering at least two compartments within the peripheral immune system: one centered on mucosal lymph nodes (referred to as compartment- μ) and another on non-mucosal lymphatic tissues draining the immunotherapeutic injection sites (compartment- η in [11]).

Based on this analysis, our model introduces three compartments: the central compartment and the peripheral compartment divided into mucosal and non-mucosal compartments.

Referring to [11], T cells initially recruited into the η -compartment will transit through lymphatic tissues in the μ -compartment and vice versa.

The mathematical modeling is performed through delayed differential equations (DDEs). Through these types of equations, we can take into account the cell migration time or the time required for the production of antibodies (which are not instantaneous). Thus, we obtain a more accurate description of the phenomena we want to model.

The paper's structure is as follows: Section 2 introduces and elaborates on the mathematical model; Section 3 explores the relevant solutions' properties; Section 4 investigates the biologically significant equilibria and their stability; Section 5 presents numerical simulations with biological interpretations; Section 6 lists the parameter values used in the simulations; and Section 7 concludes the article.

2. The Model Establishment

The model we propose comprises 10 delay differential equations, building upon the models presented in [12–15]. A notable difference lies in our consideration of three body compartments and the introduction of three delays to accurately capture the biological timeframe.

The initial four equations depict the dynamics within the non-mucosal compartment (η -compartment) following allergen entry, describing the concentrations of naive T cells, Th1, Th2, and T_{reg} cells denoted as N_η , $T_{1\eta}$, $T_{2\eta}$, and $T_{r\eta}$, respectively.

Subsequently, Equations (5)–(7) detail the behavior within the mucosal compartment (μ -compartment) concerning the concentrations of Th1, Th2, and T_{reg} cells ($T_{1\mu}$, $T_{2\mu}$, $T_{r\mu}$).

Equations (8) and (9) elucidate a two-stage maturation process of APCs post-allergen contact, where A_1 signifies the concentration of naive APCs and A_2 represents mature APCs. The final equation characterizes the production of IL-6, denoted by I .

Therefore, the model is as follows:

$$\begin{aligned}
 \dot{N}_\eta &= \alpha - \gamma_3 N_\eta - N_\eta A_2(t - \tau_2) \left(\frac{T_{1\eta}}{1 + \mu_2 T_{2\eta}} \right) - \\
 &\quad - \phi N_\eta A_2(t - \tau_2) T_{2\eta} - \kappa N_\eta A_2(t - \tau_2) T_{r\eta} \\
 \dot{T}_{1\eta} &= -(1 + \theta) T_{1\eta} + \theta T_{1\mu}(t - \tau_3) + \\
 &\quad + \frac{v N_\eta A_2(t - \tau_2)}{(1 + \mu_r T_{r\eta})} \left(\frac{T_{1\eta}}{1 + \mu_2 T_{2\eta}} \right) - \eta_1 \frac{IT_{1\eta}}{1 + I} \\
 \dot{T}_{2\eta} &= -(1 + \theta) T_{2\eta} + \theta T_{2\mu}(t - \tau_3) + \\
 &\quad + \phi \frac{v N_\eta A_2(t - \tau_2)}{(1 + \mu_r T_{r\eta})} \left(\frac{T_{2\eta}}{1 + \mu_1 \frac{T_{1\eta}}{1 + \mu_2 T_{2\eta}}} \right) + \eta_2 \frac{IT_{2\eta}}{1 + I} \\
 \dot{T}_{r\eta} &= -(1 + \theta) T_{r\eta} + \theta T_{r\mu}(t - \tau_3) + \kappa v N_\eta A_2(t - \tau_2) T_{r\eta} - \eta_r \frac{IT_{r\eta}}{1 + I} \\
 \dot{T}_{1\mu} &= -(1 + \theta) T_{1\mu} + \theta T_{1\eta}(t - \tau_3) \\
 \dot{T}_{2\mu} &= -(1 + \theta) T_{2\mu} + \theta T_{2\eta}(t - \tau_3) \\
 \dot{T}_{r\mu} &= -(1 + \theta) T_{r\mu} + \theta T_{r\eta}(t - \tau_3) \\
 \dot{A}_1 &= \lambda - \gamma_1 A_1 - \beta \Lambda A_1 \\
 \dot{A}_2 &= \beta \Lambda A_1 - \gamma_2 A_2 - \mu A_2 T_{r\eta} \\
 \dot{I} &= -\gamma_4 I + k_1 [A_2(t - \tau_1) + N_\eta(t - \tau_1) + T_{1\eta}(t - \tau_1) + \\
 &\quad + T_{2\eta}(t - \tau_1) + T_{r\eta}(t - \tau_1)]
 \end{aligned} \tag{1}$$

The initial equation represents the variation in the concentration of naive T cells characterized by a constant production rate α and a constant death rate γ_3 . The last three terms signify the differentiation of naive cells into Th1, Th2, and T_{reg} cells upon encountering mature APCs (A_2).

The subsequent three equations illustrate the evolution of the concentrations of Th1, Th2, and T_{reg} in the η -compartment. Following [14], these concentrations are related to naive cells, the presented allergen concentration, and their respective cytokines.

In each equation, the first two terms account for migration between the μ - and η -compartments, cell degradation, and the third term indicates the differentiation of naive cells into Th1, Th2, and T_{reg} . Notably, for Th1, differentiation is suppressed by T_{reg} and Th2 cells, while for Th2, suppression occurs from T_{reg} and Th1 cells. The last term in each

equation represents the effect of IL_6 during chemotherapy with desensitization dose Λ . Additionally, the equations for $Th1$ and T_{reg} incorporate inhibition rates η_1 and η_r , while the equation for $Th2$ includes a stimulation rate η_2 [6,16,17].

Parameters like v determine the proliferation rate of differentiated T cells from a single naive cell, while ϕ and κ account for autocrine action differences among mature T helper cell subsets. The suppression strengths of $Th1$, $Th2$, and T_{reg} are controlled by parameters μ_1 , μ_2 , and μ_r .

Equations (5)–(7) model concentrations of $Th1$, $Th2$, and T_{reg} cells in the μ -compartment, considering migration to and from the η -compartment with migration probability (θ).

Equations (8) and (9) portray the progression of naive and mature APCs post-allergen contact. The chemotherapeutic drug-induced allergen Λ remains constant. According to [13], T_{reg} cells play a regulatory role in limiting APC maturation.

Naive APCs exhibit a constant production rate λ and a constant death rate γ_1 . The third term in Equation (9) represents APC activation by the antigen.

Equation (10) illustrates the production of IL-6, where mature APCs and mature T-cells, following [4], contribute to its production. The first term denotes the death rate of IL_6 cells, while the second term depicts I cell production by mature APCs, Naive T cells, Th1 cells, Th2 cells, and Treg cells.

To ensure biological accuracy, three delays are introduced as follows:

- τ_1 is the time necessary for the production of IL-6;
- τ_2 is the allergen's propagation time from the central compartment to the peripheral compartment [18];
- τ_2 is the transit time due to the inter-compartmental migration of Th1, Th2 and Treg cells between the mucosal and non-mucosal compartment. The delay is prescribed by the probability of migration (k_m):

$$\tau_3 = \frac{1}{k_m}$$

A comprehensive list detailing the model's parameters and their descriptions is available in Section 6.

Introducing New Notations for State Variables

Initially, the model is introduced using common notations frequently found in the literature, aiming for ease of comprehension from a biological perspective. To facilitate the mathematical analysis of the delay differential equation (DDE) system, subsequently, the following notations are introduced:

- x_1 = concentration of naive T-cells (N_η).
- x_2 = concentration of Th1 cells in η -compartment.
- x_3 = concentration of Th2 cells in η -compartment.
- x_4 = concentration of T_{reg} cells in η -compartment.
- x_5 = concentration of Th1 cells in μ -compartment.
- x_6 = concentration of Th2 cells in μ -compartment.
- x_7 = concentration of T_{reg} cells in μ -compartment.
- x_8 = concentration of naive APCs.
- x_9 = concentration of mature APCs.
- x_{10} = concentration of IL-6.

Thus, (1) becomes:

$$\begin{aligned}
 \dot{x}_1 &= \alpha - \gamma_3 x_1 - x_1 x_{9\tau_2} \frac{x_2}{1 + \mu_2 x_3} - \phi x_1 x_{9\tau_2} x_3 - \kappa x_1 x_{9\tau_2} x_4 \\
 \dot{x}_2 &= -(1 + \theta) x_2 + \theta x_{5\tau_3} + v \frac{x_1 x_{9\tau_2} x_2}{(1 + \mu_r x_4)(1 + \mu_2 x_3)} - \eta_1 \frac{x_{10} x_2}{1 + x_{10}} \\
 \dot{x}_3 &= -(1 + \theta) x_3 + \theta x_{6\tau_3} + \phi v \frac{x_1 x_{9\tau_2} x_3 (1 + \mu_2 x_3)}{(1 + \mu_r x_4)(1 + \mu_1 x_2 + \mu_2 x_3)} + \eta_2 \frac{x_{10} x_3}{1 + x_{10}} \\
 \dot{x}_4 &= -(1 + \theta) x_4 + \theta x_{7\tau_3} + \kappa v x_1 x_{9\tau_2} x_4 - \eta_r \frac{x_{10} x_4}{1 + x_{10}} \\
 \dot{x}_5 &= -(1 + \theta) x_5 + \theta x_{2\tau_3} \\
 \dot{x}_6 &= -(1 + \theta) x_6 + \theta x_{3\tau_3} \\
 \dot{x}_7 &= -(1 + \theta) x_7 + \theta x_{4\tau_3} \\
 \dot{x}_8 &= \lambda - \gamma_1 x_8 - \beta \Lambda x_8 \\
 \dot{x}_9 &= \beta \Lambda x_8 - \gamma_2 x_9 - \mu x_9 x_4 \\
 \dot{x}_{10} &= -\gamma_4 x_{10} + k_1 (x_{9\tau_1} + x_{1\tau_1} + x_{2\tau_1} + x_{3\tau_1} + x_{4\tau_1})
 \end{aligned} \tag{2}$$

3. General Properties of the Solutions

Because the model describes a biological mechanism, a critical step is to demonstrate the positivity of the solutions.

Define $\tau = \max\{\tau_1, \tau_2, \tau_3\}$ and let $PC([-\tau, 0], \mathbb{R}^{10})$ denote the space of piecewise continuous functions defined on $[-\tau, 0]$ with values in $\mathbb{R}^n$. The norm in $PC([-\tau, 0], \mathbb{R}^{10})$ will be defined by

$$\|\varphi\|_\tau = \sup\{\|\varphi(t)\|_2; t \in [-\tau, 0]\}$$

with $\|\cdot\|_2$ the Euclidean norm in $\mathbb{R}^n$. For (2), consider the initial data

$$x(s) = \varphi(s), s \in [-\tau, 0] \tag{3}$$

Proposition 1. *If the initial data $\varphi \in PC([-\tau, 0], \mathbb{R}^{10})$ satisfy $\varphi_j(s) > 0$ for $s \in [-\tau, 0]$, $j = 1, \dots, 10$, then the solution to the Cauchy problem (2) + (3) will fulfill $x_j(t) \geq 0$ for all t within the domain of existence.*

Proof. Suppose $x_j(t) > 0 \forall t \in [-\tau, t_0]$. Then, if $x_1(t_0) = 0$, it follows that $x'_1(t_0) = \alpha > 0$ so x_1 will increase for $t > t_0$. If $x_2(t_0) = 0$, then $x'_2(t_0) = \theta x_{5\tau_3} > 0$ so x_2 increases for $t > t_0$. The same reasoning applies to $x_3, \dots, x_{10}$. $\square$

From now on, the initial data for (2) will be supposed positive.

Proposition 2. *x_1, x_8, x_9 are bounded on the whole interval of existence.*

Proof. From (2), it follows that

$$\dot{x}_1(t) = \alpha - \gamma_3 x_1(t) - x_1(t)p(t).$$

with $p(t) \geq 0$ for positive initial data. Then,

$$x_1(t) = x_1(0)e^{-\gamma_3 t - \int_0^t p(s)ds} + \alpha \left(\int_0^t e^{\gamma_3 s} e^{\int_0^s p(r)dr} ds \right) e^{-\gamma_3 t} e^{-\int_0^t p(s)ds}.$$

The following estimation for the second term holds:

$$\begin{aligned} & \alpha \left(\int_0^t e^{\gamma_3 s} e^{\int_0^s p(r) dr} ds \right) e^{-\gamma_3 t} e^{-\int_0^t p(s) ds} \leq \\ & \leq \alpha \left(\int_0^t e^{\gamma_3 s} e^{\int_0^s p(r) dr} ds \right) e^{-\gamma_3 t} e^{-\int_0^t p(s) ds} = \frac{\alpha}{\gamma_3} (1 - e^{-\gamma_3 t}) \leq \frac{\alpha}{\gamma_3}, \forall t \geq 0. \end{aligned}$$

It follows that $|x_1(t)| \leq M_1$ for some positive M_1 .

It is elementary to see that

$$x_8(t) = (x_8(0) - \frac{\lambda}{\gamma_1 + \beta\Lambda}) e^{-(\gamma_1 + \beta\Lambda)t} + \frac{\lambda}{\gamma_1 + \beta\Lambda},$$

so $|x_8(t)| \leq M_8, \forall t \in [-\tau, \infty)$ for some $M_8 > 0$.

Concerning x_9 , the variation of constants formula gives

$$x_9(t) = x_9(0) e^{-\gamma_2 t - \mu \int_0^t x_4(s) ds} + \beta\Lambda \left(\int_0^t x_8(s) e^{\gamma_2 s + \mu \int_0^s x_4(r) dr} ds \right) e^{-\gamma_2 t - \mu \int_0^t x_4(s) ds}.$$

The first term is obviously bounded for $t \geq 0$. For the second one, we have the following estimations:

$$\begin{aligned} & \beta\Lambda \left| \int_0^t x_{8\tau 1}(s) e^{\gamma_2 s + \mu \int_0^s x_4(r) dr} ds \right| e^{-\gamma_2 t - \mu \int_0^t x_4(s) ds} \leq \\ & \leq \beta\Lambda \left(\int_0^t |x_8(s)| e^{\gamma_2 s + \mu \int_0^s x_4(r) dr} ds \right) e^{-\gamma_2 t - \mu \int_0^t x_4(s) ds} \leq \\ & \leq \beta\Lambda M_8 \left(\int_0^t e^{\gamma_2 s + \mu \int_0^s x_4(r) dr} ds \right) e^{-\gamma_2 t} e^{-\int_0^t x_4(s) ds} = \\ & = \beta\Lambda M_8 \frac{e^{\gamma_2 t} - 1}{\gamma_2} e^{-\gamma_2 t} \leq \frac{\beta\Lambda M_8}{\gamma_2}, \end{aligned}$$

so $|x_9(t)| \leq M_9 \forall t \in [-\tau, \infty)$ with $M_9 > 0$. $\square$

Proposition 3. The solution of (2) exists on $[-\tau, \infty)$.

Proof. The proposition follows from a slight generalization of Theorem 1.2. in [19]. Namely, the condition imposed to the right-hand side of the equation should hold only for the solutions of the equation instead of all functions from $PC[-\tau, 0]$. So we must show that, with $\varphi = (\varphi_1, \dots, \varphi_{10})$, a solution of (2) and $f = (f_1, \dots, f_{10})$, the right-hand side of (2),

$$|f_j(\varphi)| \leq h(\|\varphi\|_\tau), \int_{r_0}^\infty \frac{1}{h(r)} dr = \infty, \forall r_0 > 0.$$

We will show that there exist constants C_1, C_2 so that

$$|f_j(\varphi)| \leq C_1 + C_2 \|\varphi\|_\tau, j = 1, \dots, 10.$$

$$|f_1(\varphi)| \leq |\alpha| + \gamma_3 |\varphi_1(t)| + M_1 M_9 |\varphi_2(t)| + \phi M_1 M_9 |\varphi_3(t)| + \kappa M_1 M_9 |\varphi_4(t)| \leq$$

$$\leq |\alpha| + (\gamma_3 + M_1 M_9 + \phi M_1 M_9 + \kappa M_1 M_9) \|\varphi\|_\tau,$$

$$|f_2(\varphi)| \leq (1 + \theta) |\varphi_2(t)| + \theta |\varphi_5(t - \tau_3)| + (v M_1 M_9 + \eta_1) |\varphi_2(t)| \leq$$

$$\leq (1 + 2\theta + v M_1 M_9 + \eta_1) \|\varphi\|_\tau,$$

$$|f_3(\varphi)| \leq (1 + \theta) |\varphi_3(t)| + \theta |\varphi_6(t - \tau_3)| + (\phi v M_1 M_9 + \eta_2) |\varphi_3(t)| \leq$$

$$\leq (1 + 2\theta + \phi v M_1 M_9 + \eta_2) \|\varphi\|_\tau,$$

$$|f_4(\varphi)| \leq (1 + \theta) |\varphi_4(t)| + \theta |\varphi_7(t - \tau_3)| + (\kappa v M_1 M_9 + \eta_r) |\varphi_4(t)| \leq$$

$$\leq (1 + 2\theta + \kappa v M_1 M_9 + \eta_r) \|\varphi\|_\tau,$$

$$|f_5(\varphi)| \leq (1 + \theta) |\varphi_5(t)| + \theta |\varphi_2(t - \tau_3)| \leq (1 + 2\theta) \|\varphi\|_\tau,$$

$$|f_6(\varphi)| \leq (1 + 2\theta) \|\varphi\|_\tau, |f_7(\varphi)| \leq (1 + 2\theta) \|\varphi\|_\tau,$$

$$|f_8(\varphi)| \leq |\lambda| + (\gamma_1 + \beta \Lambda) \|\varphi\|_\tau,$$

$$|f_9(\varphi)| \leq (\beta \Lambda + \gamma_2 + \mu M_9) \|\varphi\|_\tau,$$

$$|f_{10}(\varphi)| \leq (\gamma_4 + 5k_1) \|\varphi\|_\tau.$$

The proposition is proved. $\square$

4. Equilibria and Stability Analysis

The system (2) possesses two equilibrium points that are particularly significant for our study:

$$E_1 = (x_1^*, 0, 0, 0, 0, 0, 0, 0, x_8^*, x_9^*, x_{10,10}^*)$$

$$E_2 = (\hat{x}_1, \hat{x}_2, 0, 0, \hat{x}_5, 0, 0, \hat{x}_8, \hat{x}_9, \hat{x}_{10})$$

Both equilibrium points hold positive implications for patients. In the scenario of E_1 , there exists no immune response to the allergen, while for E_2 , the immune response lacks association with allergies, evidenced by $x_2 > x_3$ ($Th1 > Th2$). The stability of these equilibrium points is pivotal for patient well-being.

Subsequently, we analyze the stability properties of these equilibrium points.

4.1. Equilibrium Point E_1

Let $E_1 = (x_1^*, 0, 0, 0, 0, 0, 0, 0, x_8^*, x_9^*, x_{10,10}^*)$ with

$$x_1^* = \alpha, x_8^* = \frac{\lambda}{\gamma_1 + \beta \Lambda}, x_9^* = \frac{\beta \Lambda \lambda}{\gamma_2 (\gamma_1 + \beta \Lambda)}, x_{10}^* = \frac{k_1 x_9^*}{\gamma_4}$$

The matrix A , comprising partial derivatives concerning the undelayed variables and computed at E_1 , exhibits the following non-zero elements:

$$\begin{aligned} a_{11} &= -\gamma_3, \\ a_{12} &= x_1^* x_9^*, \\ a_{13} &= -\phi x_1^* x_9^*, \\ a_{14} &= -k x_1^* x_9^*, \\ a_{22} &= -(1 + \theta) + v x_1^* x_9^* - \eta_1 \frac{x_{10}^*}{1 + x_{10}^*}, \\ a_{33} &= -(1 + \theta) + \phi x_1^* x_9^* + \eta_2 \frac{x_{10}^*}{1 + x_{10}^*}, \\ a_{44} &= -(1 + \theta) + k v x_1^* x_9^* - \eta_r \frac{x_{10}^*}{1 + x_{10}^*}, \\ a_{55} &= a_{66} = a_{77} = -(1 + \theta), \\ a_{88} &= -\gamma_1 - \beta \Lambda, \\ a_{94} &= -\mu x_9^*, \\ a_{98} &= \beta \Lambda, \\ a_{99} &= -\gamma_2, \\ a_{10,10} &= -\gamma_4. \end{aligned}$$

The matrix A_{τ_1} of partial derivatives concerning the τ_1 -delayed variables contains the only non-zero elements.

$$b_{10,1} = b_{10,2} = b_{10,3} = b_{10,4} = b_{10,9} = k_1 \text{ \& } b_{10,10} = -\gamma_4.$$

The matrix A_{τ_2} pertaining to partial derivatives concerning the τ_2 -delayed variables is zero. Lastly, the matrix A_{τ_3} , representing partial derivatives concerning the τ_3 -delayed variables, contains the nonzero elements:

$$d_{25} = d_{36} = d_{47} = d_{52} = d_{63} = d_{74} = \theta.$$

The characteristic equation for E_1 is

$$\det(\lambda I_{10} - A - \sum_{j=1}^3 A_{\tau_j} e^{\lambda \tau_j}) = 0 \Leftrightarrow (\lambda - a_{10,10})(\lambda - a_{99})(\lambda - a_{88})(\lambda - a_{11})d_1(\lambda) = 0$$

$$d_1(\lambda) = \begin{vmatrix} \lambda - a_{22} & 0 & 0 & -\theta e^{-\lambda \tau_3} & 0 & 0 \\ 0 & \lambda - a_{33} & 0 & 0 & -\theta e^{-\lambda \tau_3} & 0 \\ 0 & 0 & \lambda - a_{44} & 0 & 0 & -\theta e^{-\lambda \tau_3} \\ -\theta e^{-\lambda \tau_3} & 0 & 0 & \lambda - a_{55} & 0 & 0 \\ 0 & -\theta e^{-\lambda \tau_3} & 0 & 0 & \lambda - a_{66} & 0 \\ 0 & 0 & -\theta e^{-\lambda \tau_3} & 0 & 0 & \lambda - a_{77} \end{vmatrix}$$

To address the challenges associated with analyzing the latter part of the characteristic equation, we will employ an alternative approach for the case where $d_1 = 0$. It is noteworthy that the equation $d_1(\lambda) = 0$ represents the characteristic equation of a linear system:

$$\dot{x} = A_1 x + B_1 x_{\tau_3} \quad (4)$$

where A_1 is diagonal, and B_1 follows the Meltzer structure (see [20]). One can then apply the criterion from [20] as presented in Proposition 2.3: If the matrix $A_1 + B_1$ is Hurwitz, and if the delay τ_3 is sufficiently small, the system (4) is exponentially stable. Consequently, it follows that if

$$a_{10,10} < 0, a_{99} < 0, a_{88} < 0, a_{11} < 0, \quad (5)$$

and this is indeed the case, then the equilibrium point E_1 is asymptotically stable.

Below, sufficient conditions ensuring that $A_1 + B_1$ is Hurwitz are provided. It is worth noting that the polynomial derived from d_1 when $\tau_3 = 0$ should be Hurwitz, meaning the roots of the corresponding equation must have negative real parts.

Now, when $\tau_3 = 0$, the equation $d_1(\lambda) = 0$ transforms into:

$$(\lambda - a_{33})(\lambda - a_{44})(\lambda - a_{66})(\lambda - a_{77}) - \theta^2[(\lambda - a_{33})(\lambda - a_{66}) + (\lambda - a_{44})(\lambda - a_{77})] + \theta^4 = 0 \quad (6)$$

To address Equation (6), we introduce

$$u_1 = (\lambda - a_{33})(\lambda - a_{66}), u_2 = (\lambda - a_{44})(\lambda - a_{77})$$

so, with $s = \theta^2$, Equation (6) becomes

$$s^2 - s(u_1 + u_2) + u_1 u_2 = 0 \quad (7)$$

with solutions $s_1 = u_1, s_2 = u_2$.

Consequently, the roots of the following equations must have real parts smaller than zero:

$$\lambda^2 - (a_{33} + a_{66})\lambda + a_{33}a_{66} - \theta^2 = 0, \lambda^2 - (a_{44} + a_{77})\lambda + a_{44}a_{77} - \theta^2 = 0. \quad (8)$$

The following necessary and sufficient conditions ensure that Equation (8) possess roots with negative real parts:

$$a_{33} + a_{66} < 0, a_{33}a_{66} - \theta^2 > 0, a_{44} + a_{77} < 0, a_{44}a_{77} - \theta^2 > 0 \quad (9)$$

The following theorem results.

Theorem 1. Assuming conditions (5) and (9) are satisfied, then, for

$$\tau_3 < \frac{1}{e|a_{jj}|}, j = 2, \dots, 7$$

the equilibrium point E_1 is asymptotically stable.

Proof. The conclusion is reached by applying Proposition 2.3. from [20] and the preceding reasoning. $\square$

4.2. Equilibrium Point E_2

Let us now consider the equilibrium point

$$E_2 = (\hat{x}_1, \hat{x}_2, 0, 0, \hat{x}_5, 0, 0, \hat{x}_8, \hat{x}_9, \hat{x}_{10}).$$

The values for E_2 are derived from the following equations:

$$\hat{x}_8 = \frac{\lambda}{\gamma_1 + \beta\Lambda}$$

$$\hat{x}_9 = \frac{\lambda\beta\Lambda}{\gamma_2(\gamma_1 + \beta\Lambda)}$$

$$\hat{x}_2 = \frac{\alpha - \gamma_3\hat{x}_1}{\hat{x}_1\hat{x}_9}$$

$$\hat{x}_{10} = \frac{k_1}{\gamma_4}(\hat{x}_9 + \hat{x}_1 + \frac{\alpha - \gamma_3\hat{x}_1}{\hat{x}_1\hat{x}_9})$$

$$\hat{x}_5 = \frac{\theta\hat{x}_2}{1 + \theta}$$

Substitute these values into the second Equation of (2) and simplify $\hat{x}_2$. Let $c_1 = \frac{1+2\theta}{1+\theta}$. The subsequent equation yields $\hat{x}_1$:

$$\begin{aligned} & k_1 v \hat{x}_9^2 x^3 + (v \gamma_4 \hat{x}_9^2 + k_1 v \hat{x}_9^3 - k_1 v \gamma_3 \hat{x}_9 - c k_1 \hat{x}_9 - \eta_1 k_1 \hat{x}_9) x^2 + \\ & + (k_1 v \alpha \hat{x}_9 - c \gamma_4 \hat{x}_9 - c k_1 \hat{x}_9^2 + c k_1 \gamma_3 - \eta_1 k_1 \hat{x}_9^2 + \eta_1 k_1 \gamma_3) x - c k_1 \alpha - \eta_1 k_1 \alpha = 0 \end{aligned} \quad (10)$$

(10) can be expressed as:

$$ax^3 + bx^2 + cx + d = 0.$$

Calculate, as previously done, the partial derivatives in E_2 , and denote

$$b_2 = \phi v \frac{\hat{x}_1 \hat{x}_9}{1 + \mu_1 \hat{x}_2} + \eta_2 \frac{\hat{x}_{10}}{1 + \hat{x}_{10}},$$

$$b_3 = \kappa v \hat{x}_1 \hat{x}_9 - \eta_r \frac{\hat{x}_{10}}{1 + \hat{x}_{10}}.$$

Revisit now the following definitions from [21]. Consider the system

$$\dot{x} = X(t, x_t), x_{t_0} = \varphi \quad (11)$$

Assume that $\tau > 0$ is a given real number, $x_t : [-\tau, 0] \rightarrow \mathbb{R}^n$ is defined by $x_t(s) = x(t+s)$, with the norm $\|x(t)\|_2 = \sqrt{x_1^2(t) + \dots + x_n^2(t)}$. Introduce the partition

$x = (z, u)^T$ (T denotes transposition), where $z \in \mathbb{R}^m$, $u \in \mathbb{R}^{n-m}$ ($1 \leq m < n$).

$X : \mathbb{R} \times C \rightarrow \mathbb{R}^n$ is assumed to be continuous and X maps every $\mathbb{R} \times$ (bounded set) into a bounded set, in the domain D defined by

$$t \geq 0, \quad \|z\|_\tau \leq h, \quad \|u\|_\tau < \infty \quad (12)$$

The solutions of (11) are assumed to be unique and uncontinuable: the solutions are defined for every $t \geq t_0$ where $\|z(t; t_0, \varphi)\|_2 \leq h$. If we suppose that (12) holds for all $t \geq t_0$, then the solution $x(t; t_0, \varphi)$ is defined for all $t \geq t_0$.

Taking into account the above partitions, the system (11) can be represented as:

$$\dot{z}(t) = Z(t, y_t, z_t), \dot{u}(t) = U(t, y_t, z_t) \quad (13)$$

Suppose that

$$Z(t, 0, 0) = 0, \quad U(t, 0, 0) = 0 \quad \forall t \geq t_0 \quad (14)$$

Let $x(t; t_0, \varphi)$ denote a solution of the system (11) with initial condition $x(t_0; \varphi)$. The same notation will be applied for the solutions of each of the two equations in (14).

Definition 1 ([21–23]). The equilibrium point $x = 0$ in system (11), (13) is called the following:

1. *z-stable*, if for every $t_0 \geq 0$ and every $\epsilon > 0$, there exists a $\delta(\epsilon; t_0) > 0$ such that $\|\varphi\|_\tau < \delta$ implies $\|z(t; t_0; \varphi)\|_2 < \epsilon$ for all $t \geq t_0$. It is called *uniformly z-stable* if δ does not depend on t_0 .
2. *Asymptotically z-stable* if it is *z-stable* and for every $t_0 \geq 0$ there exists a $\Delta(t_0) > 0$ such that for every solution $x(t; t_0; \varphi)$ of system (11), (13) that satisfies $\|\varphi\|_\tau < \Delta(t_0)$, the following holds true:

$$\lim_{t \rightarrow \infty} \|z(t; t_0; \varphi)\| = 0 \quad (15)$$

3. *Uniformly asymptotically z-stable*, if it is *uniformly z-stable* with respect to t_0 in terms of point 1, and one can find $\Delta > 0$ such that relation (15) is met uniformly with respect to (t_0, φ) from the domain $t_0 \geq 0, \|\varphi\|_\tau < \Delta$ (for any numbers $\eta > 0, t_0 \geq 0$ one can find the number $T = T(\eta) > 0$ such that $\|z(t; t_0, \varphi)\| < \eta$ for all $t \geq t_0 + T, \|\varphi\|_\tau < \Delta$).

The following theorem will be used to prove the stability of E_2 with respect to some of its variables.

Theorem 2 ([21], Th.5.2.1). Suppose there exists a function $V : \mathbb{R} \times C \rightarrow \mathbb{R}$ of class C^1 and continuously increasing functions.

$a, b : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ with $a(0) = b(0) = 0$ so that, in the domain (12),

$$V(t, \varphi) \geq a(\|\varphi_v(0)\|), \quad \dot{V}(t, x_t) \leq 0. \quad (16)$$

where $\dot{V}$ denotes the derivative along system (11), then the equilibrium point $x = 0$ of system (11) is z -stable.

If, in addition,

$$V(t, \varphi) \leq b(\|\varphi\|_\tau), \quad (17)$$

then the z -stability is uniform.

Theorem 3. Suppose $b_2 < 1$, $b_3 < 1$ and $\frac{\beta\Lambda}{2} < \gamma_2$. Then the equilibrium point E_2 exhibits partial stability concerning the variables $(x_1, x_3, x_4, x_6, x_7, x_8, x_9)$.

Proof. Shift the equilibrium to zero by utilizing $z_i = x_i - \hat{x}_i$, where $i = 1, \dots, 10$ (assuming $\hat{x}_3 = 0, \hat{x}_4 = 0, \hat{x}_6 = 0, \hat{x}_7 = 0$). Introduce $z = (z_1, \dots, z_{10})$, and define

$$V(z) = \sum_{i=3}^4 \left(\frac{z_i^2}{2} + \frac{\theta}{2} \int_{t-\tau_3}^t z_i(s)^2 ds \right) + \sum_{i=6}^7 \left(\frac{z_i^2}{2} + \frac{\theta}{2} \int_{t-\tau_3}^t z_i(s)^2 ds \right) + \frac{z_1^2}{2} + \frac{z_8^2}{2} + \frac{z_9^2}{2}$$

Denote as $y = (z_1, z_3, z_4, z_6, z_7, x_8, x_9) = (x_1, x_3, x_4, x_6, x_7, x_8, x_9)$. Additionally, it should be noted that the initial condition to establish a (partial) Lyapunov–Krasovskii functional is confirmed by V :

$$V(z) \geq v(\|y\|_2).$$

Also,

$$V(z) \leq M\|z\|_\tau^2$$

for some $M_1 > 0$. The derivative of V along the adjusted system (11) is expressed as:

$$\begin{aligned} \frac{dV}{dt} &= z_3 f_3 + z_4 f_4 + z_6 f_6 + z_7 f_7 + \frac{\theta}{2} (z_3^2 - z_{3\tau_3}^2 + z_4^2 - z_{4\tau_3}^2 + z_6^2 - z_{6\tau_3}^2 + z_7^2 - z_{7\tau_3}^2) \\ &+ z_1 f_1 + z_8 f_8 + z_9 f_9 \\ &= -(1 - b_2) z_3^2 - (1 - b_3) z_4^2 - (1 + \theta) z_6^2 - (1 + \theta) z_7^2 - \frac{\theta}{2} (z_3^2 - 2z_3 z_{6\tau_3} + z_{6\tau_3}^2) - (1 + \theta) z_5^2 \\ &- \frac{\theta}{2} (z_4^2 - 2z_4 z_{7\tau_3} + z_{7\tau_3}^2) - \frac{\theta}{2} (z_6^2 - 2z_6 z_{3\tau_3} + z_{3\tau_3}^2) - \frac{\theta}{2} (z_7^2 - 2z_7 z_{4\tau_3} \\ &+ z_{4\tau_3}^2) + z_3^2 \left[\frac{\phi v(z_1 + \hat{x}_1)(z_9 \tau_2 + \hat{x}_9)(1 + \mu_2 z_3)}{(1 + \mu_r z_4)(1 + \mu_2 z_3 + \mu_1(z_2 + \hat{x}_2))} - \phi v \frac{\hat{x}_1 \hat{x}_9}{1 + \mu \hat{x}_2} + \eta_2 \frac{z_{10}}{(1 + z_{10} + \hat{x}_{10})(1 + \hat{x}_{10})} \right] \\ &+ z_4^2 \left[\kappa v(z_1 + \hat{x}_1)(z_9 \tau_2 + \hat{x}_9) - \eta_r \frac{z_{10} + \hat{x}_{10}}{1 + z_{10} + \hat{x}_{10}} - \kappa v \hat{x}_1 \hat{x}_9 + \eta_r \frac{\hat{x}_{10}}{(1 + \hat{x}_{10})} \right] \\ &- \gamma_3 z_1^2 - (z_1 + \hat{x}_1)(z_9 \tau_2 + \hat{x}_9) \frac{(z_2 + \hat{x}_2) z_1}{1 + \mu_2 z_3} - \phi(z_1 + \hat{x}_1)(z_9 \tau_2 + \hat{x}_9) z_3 z_1 - \kappa(z_1 + \hat{x}_1^*)(z_9 \tau_2 + \hat{x}_9) z_4 z_1 \\ &- (\gamma_1 + \beta\Lambda) z_8^2 + \beta\Lambda z_8 z_9 - \gamma_2 z_9^2 - \mu(z_9 + \hat{x}_9) z_4 z_9 \end{aligned}$$

By removing some negative terms and using $ab < \frac{a^2}{2} + \frac{b^2}{2}$, we obtain

$$\begin{aligned}
\frac{dV}{dt} &< -(1-b_2)z_3^2 - (1-b_3)z_4^2 - (1+\theta)z_6^2 - (1+\theta)z_7^2 - \frac{\theta}{2}(z_3^2 - 2z_3z_{6\tau_3} + z_{6\tau_3}^2) - (1+\theta)z_5^2 \\
&- \frac{\theta}{2}(z_4^2 - 2z_4z_{7\tau_3} + z_{7\tau_3}^2) - \frac{\theta}{2}(z_6^2 - 2z_6z_{3\tau_3} + z_{3\tau_3}^2) - \frac{\theta}{2}(z_7^2 - 2z_7z_{4\tau_3} \\
&+ z_{4\tau_3}^2) + z_3^2 \left[\frac{\phi v(z_1 + \hat{x}_1)(z_{9\tau_2} + \hat{x}_9)(1 + \mu_2 z_3)}{(1 + \mu_r z_4)(1 + \mu_2 z_3 + \mu_1(z_2 + \hat{x}_2))} - \phi v \frac{\hat{x}_1 \hat{x}_9}{1 + \mu \hat{x}_2} + \eta_2 \frac{z_{10}}{(1 + z_{10} + \hat{x}_{10})(1 + \hat{x}_{10})} \right] \\
&+ z_4^2 \left[\kappa v(z_1 + \hat{x}_1)(z_{9\tau_2} + \hat{x}_9) - \eta_r \frac{z_{10} + \hat{x}_{10}}{1 + z_{10} + \hat{x}_{10}} - \kappa v \hat{x}_1 \hat{x}_9 + \eta_r \frac{\hat{x}_{10}}{(1 + \hat{x}_{10})} \right] \\
&- \gamma_3 z_1^2 - (z_1 + \hat{x}_1)(z_{9\tau_2} + \hat{x}_9) \frac{(z_2 + \hat{x}_2)z_1}{1 + \mu_2 z_3} - \phi(z_1 + \hat{x}_1)(z_{9\tau_2} + \hat{x}_9)z_3 z_1 - \kappa(z_1 + \hat{x}_1^*)(z_{9\tau_2} + \hat{x}_9)z_4 z_1 \\
&- (\gamma_1 + \beta\Lambda)z_8^2 + \beta\Lambda z_8 z_9 - \gamma_2 z_9^2 - \mu(z_9 + \hat{x}_9)z_4 z_9 \\
&= -(1-b_2)z_3^2 - (1-b_3)z_4^2 - (1+\theta)z_6^2 - (1+\theta)z_7^2 - \frac{\theta}{2}(z_3^2 - 2z_3z_{6\tau_3} + z_{6\tau_3}^2) - (1+\theta)z_5^2 \\
&- \frac{\theta}{2}(z_4^2 - 2z_4z_{7\tau_3} + z_{7\tau_3}^2) - \frac{\theta}{2}(z_6^2 - 2z_6z_{3\tau_3} + z_{3\tau_3}^2) - \frac{\theta}{2}(z_7^2 - 2z_7z_{4\tau_3} \\
&+ z_{4\tau_3}^2) + z_3^2 \left[\frac{\phi v(z_1 + \hat{x}_1)(z_{9\tau_2} + \hat{x}_9)(1 + \mu_2 z_3)}{(1 + \mu_r z_4)(1 + \mu_2 z_3 + \mu_1(z_2 + \hat{x}_2))} - \phi v \frac{\hat{x}_1 \hat{x}_9}{1 + \mu \hat{x}_2} + \eta_2 \frac{z_{10}}{(1 + z_{10} + \hat{x}_{10})(1 + \hat{x}_{10})} \right] \\
&+ z_4^2 \left[\kappa v(z_1 + \hat{x}_1)(z_{9\tau_2} + \hat{x}_9) - \eta_r \frac{z_{10} + \hat{x}_{10}}{1 + z_{10} + \hat{x}_{10}} - \kappa v \hat{x}_1 \hat{x}_9 + \eta_r \frac{\hat{x}_{10}}{(1 + \hat{x}_{10})} \right] \\
&- \gamma_3 z_1^2 - (z_1 + \hat{x}_1)(z_{9\tau_2} + \hat{x}_9) \frac{(z_2 + \hat{x}_2)z_1}{1 + \mu_2 z_3} - \phi(z_1 + \hat{x}_1)(z_{9\tau_2} + \hat{x}_9)z_3 z_1 - \kappa(z_1 + \hat{x}_1^*)(z_{9\tau_2} + \hat{x}_9)z_4 z_1 \\
&- (\gamma_1 + \beta\Lambda)z_8^2 + \frac{\beta\Lambda}{2}z_8^2 + \frac{\beta\Lambda}{2}z_9^2 - \gamma_2 z_9^2 - \mu(z_9 + \hat{x}_9)z_4 z_9 \\
&< -(1-b_2)z_3^2 - (1-b_3)z_4^2 - (1+\theta)z_6^2 - (1+\theta)z_7^2 - \frac{\theta}{2}(z_3^2 - 2z_3z_{6\tau_3} + z_{6\tau_3}^2) - (1+\theta)z_5^2 \\
&- \frac{\theta}{2}(z_4^2 - 2z_4z_{7\tau_3} + z_{7\tau_3}^2) - \frac{\theta}{2}(z_6^2 - 2z_6z_{3\tau_3} + z_{3\tau_3}^2) - \frac{\theta}{2}(z_7^2 - 2z_7z_{4\tau_3} \\
&+ z_{4\tau_3}^2) + z_3^2 \left[\frac{\phi v(z_1 + \hat{x}_1)(z_{9\tau_2} + \hat{x}_9)(1 + \mu_2 z_3)}{(1 + \mu_r z_4)(1 + \mu_2 z_3 + \mu_1(z_2 + \hat{x}_2))} - \phi v \frac{\hat{x}_1 \hat{x}_9}{1 + \mu \hat{x}_2} + \eta_2 \frac{z_{10}}{(1 + z_{10} + \hat{x}_{10})(1 + \hat{x}_{10})} \right] \\
&+ z_4^2 \left[\kappa v(z_1 + \hat{x}_1)(z_{9\tau_2} + \hat{x}_9) - \eta_r \frac{z_{10} + \hat{x}_{10}}{1 + z_{10} + \hat{x}_{10}} - \kappa v \hat{x}_1 \hat{x}_9 + \eta_r \frac{\hat{x}_{10}}{(1 + \hat{x}_{10})} \right] \\
&- \gamma_3 z_1^2 - (\gamma_1 + \frac{\beta\Lambda}{2})z_8^2 + (\frac{\beta\Lambda}{2} - \gamma_2)z_9^2 \\
&= -\omega(\|z_t\|_\tau^2) + F(z_t)
\end{aligned}$$

with ω being strictly positively defined. It is important to note that, according to Propositions 1 and 2, the terms multiplying x_3^2 and x_4^2 in F are bounded, and therefore,

$$|F(z_t)| \leq C\|z_t\|_\tau^3$$

for some positive C . Therefore, the derivative of V along system (3) is strictly negatively defined for $\|z_t\| < 1$, establishing uniform partial stability as mentioned (see [21–26]). $\square$

Remark 1. The same reasoning can be applied to demonstrate partial stability for the equilibrium points where $x_2 = x_3 = x_5 = x_6 = 0$ and $x_2 = x_4 = x_5 = x_7 = 0$.

Remark 2. The partial asymptotic stability extends to x_8 through direct computation of the solution's behavior.

5. Numerical Simulations

The numerical simulations were obtained with the values for the parameters given in Section 6. These were taken from the existing literature.

When the stability for E_1 holds, we will have a successful therapy, and it shows that small quantities of allergens do not cause harm. Fortunately, using the above mentioned parameter values, we can show that the equilibrium point E_1 is stable (see Figure 1).

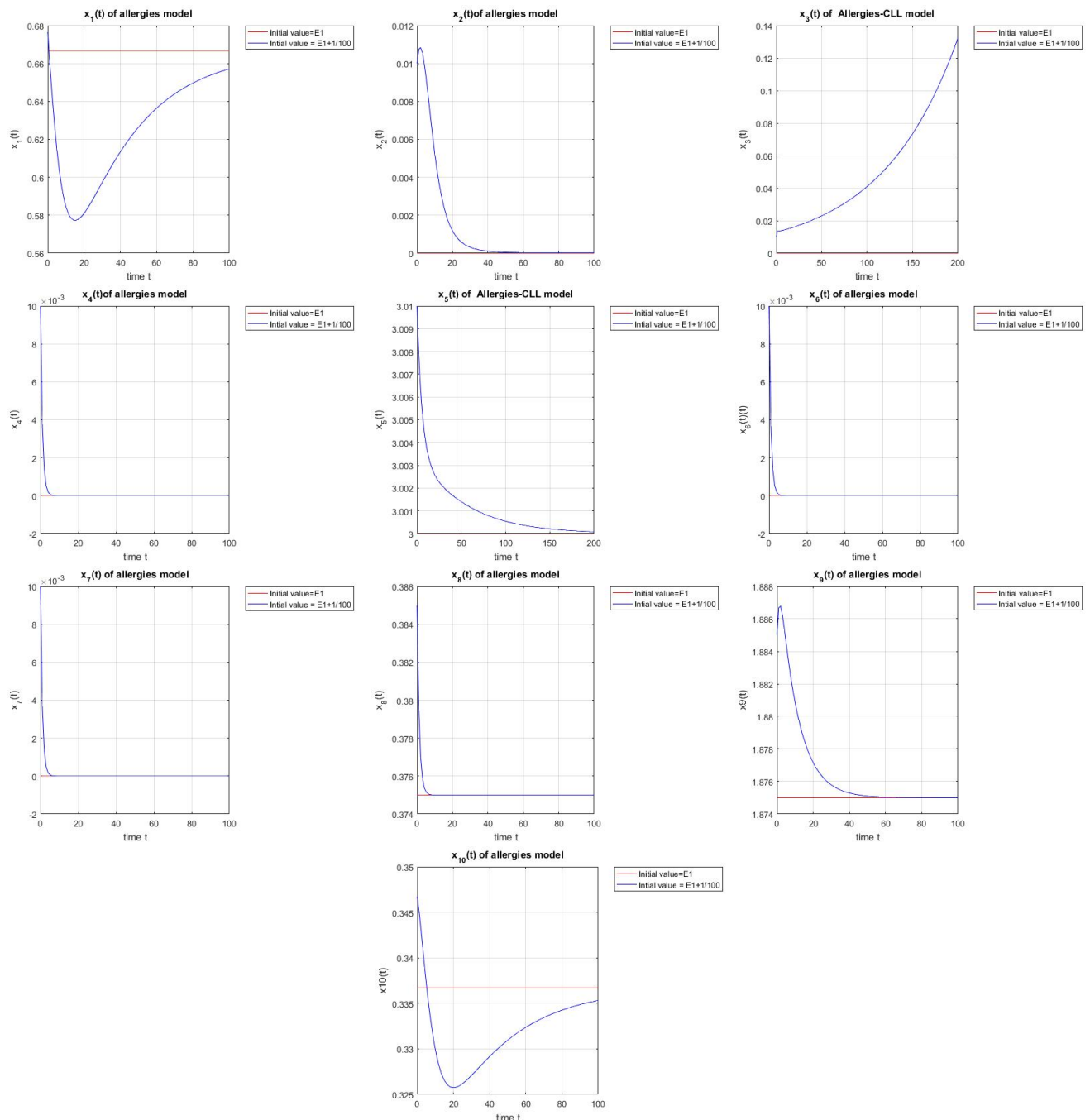


Figure 1. For a small disturbance in the initial conditions near E_1 , the system exhibits stability. Here, $E_1 = [0.6667, 0, 0, 0, 0, 0, 0, 0.3750, 1.8750, 0.3367]$.

When the stability for E_2 holds, we will have successful immunotherapy of allergies induced by chemotherapy because the Th1 cell population will dominate Th2 cell population (see Figures 2 and 3).

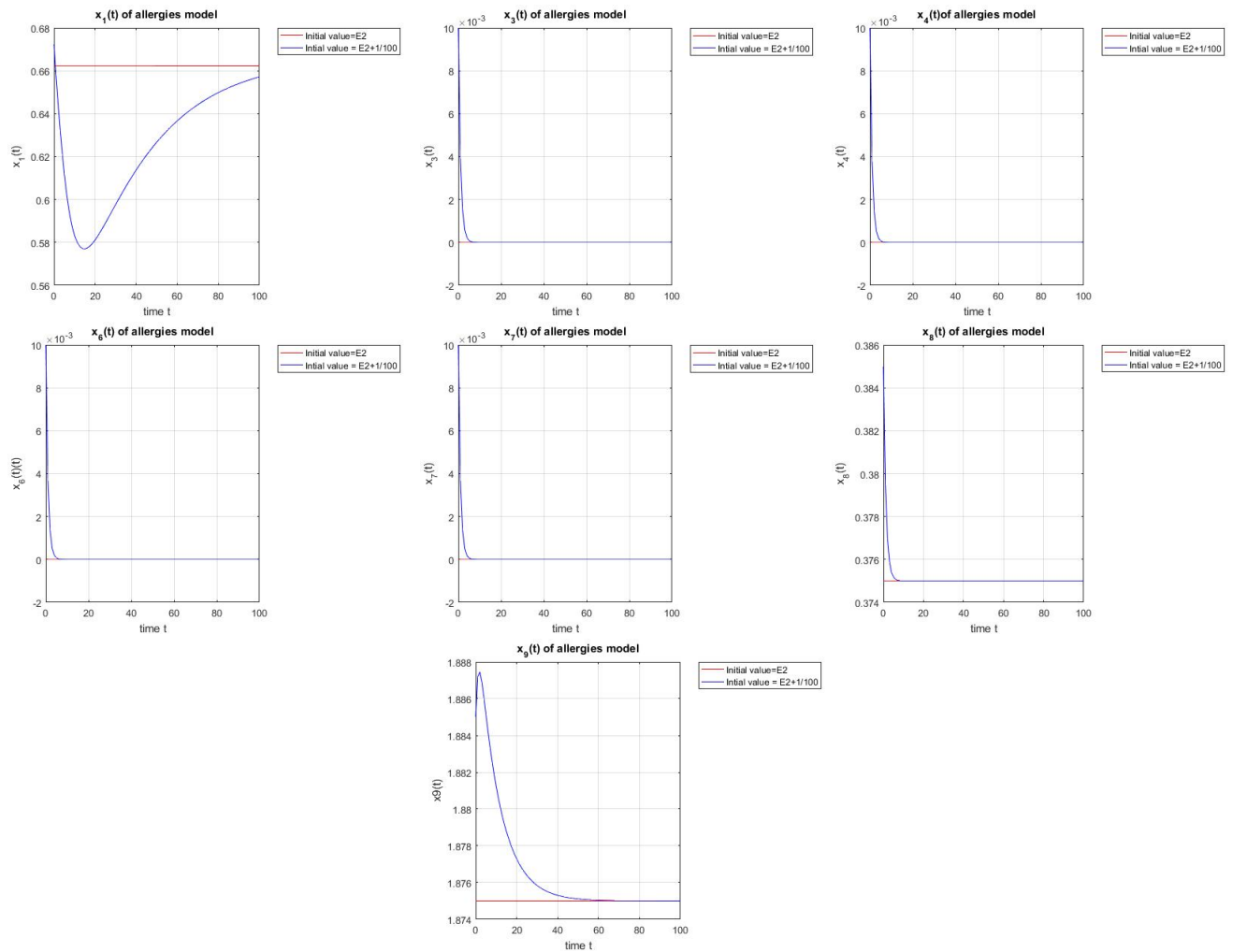


Figure 2. The stable variables in E_2 as proved in Theorem 3.

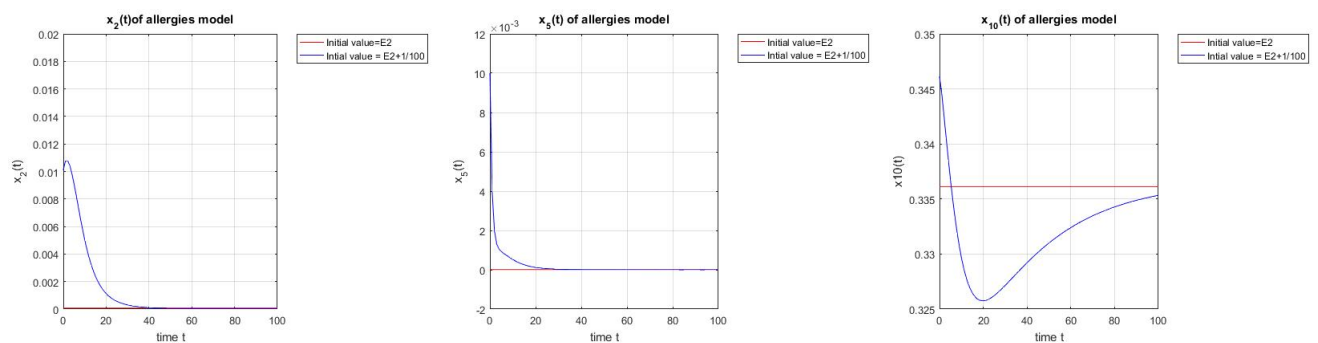


Figure 3. The remaining variables in E_2 . $E_2 = [0.6623, 1.0549 \times 10^{-4}, 0, 0, 9.5901 \times 10^{-6}, 0, 0, 0.3750, 1.8750, 0.3361]$.

6. List of Parameters

The numerical values for the system's parameters were taken from the relevant literature (see Table 1).

Table 1. Parameter values.

The production rate of naive cells [27]	α	0.02
The proliferation rate of stimulated T cells [14]	v	1
Cytokines produced by the immune system [14]	c	10^{-4}
The strength of the suppression rate of Th1 by Th2 [14]	μ_2	0.1
The strength of suppression of Th2 by Th1 [14]	μ_1	0.2
The strength of the suppression rate by Treg [14]	μ_r	0.25
The differences in the autocrine action of the three subsets in Equation (3) [14]	ϕ	0.05
The differences in the autocrine action of the three subsets in Equation (4) [14]	κ	0.1
The death rate of naive APCs [27]	γ_1	0.3
The death rate of mature APCs [27]	γ_2	0.1
The death rate of naive T cells [27]	γ_3	0.03
The natural decay of IL-6 [9]	γ_4	0.4152
The inhibition rate of Th1 cells by IL-6 [28]	η_1	0.60
The stimulation rate of Th2 cells by IL-6 [28]	η_2	0.007
The inhibition rate of Treg cells by IL-6 [28]	η_r	0.39
The amount of the injected drug dose during desensitization [13]	Λ	1
Delay to production of IL-6 [28]	τ_1	0.25
The second time delay [18]	τ_2	0.0794
The third time delay [11]	τ_3	0.08
The inter compartment migration rate [14]	θ	0.1
The production of IL-6 by other cells [29]	k_1	0.055
The birth rate of naive APCs [27]	λ	0.3
The rate of APC activation by the antigen [13]	β	0.5
The rate of APC inhibition by regulatory T cells [13]	μ	10^{-2}

7. Conclusions

We proposed a mathematical model which describes the evolution of T helper cells, APCs and IL-6 in the case of drug allergies with exposure to drug desensitization during chemotherapy. The novelty of the model consists in a more detailed approach of the biological mechanism. We considered a partition of the body into a central compartment and a peripheral one, where the latter is comprised of the mucosal and the non-mucosal compartments.

The separation of the mucosal and non-mucosal compartments is biologically accurate. It influences the evolution of the cell population, as the migration impacts the number of cells remaining in the non-mucosal compartment.

Another important improvement from the existing literature is the presence of time delays. These help to better capture the correct timeframe of the biological processes. We considered three delays which model the time necessary for the production and migration of different cell populations.

From a mathematical point of view, we studied the existence, boundedness and positivity of solutions. The positivity of the solutions was an essential study to undertake, as we are modeling cell populations. We also gave parameter conditions for stability in the case of some equilibrium points. The stability of the equilibrium points in question illustrate the benefits of drug desensitization. The numerical simulations we performed validate the theoretical results, basically showing that a small amount of allergen does not cause any harm in therapy.

Simply put, this paper introduces a delay differential equations model that describes the cellular changes that occur in the action of the immune system, in the presence of an allergen. The biological insights provided by the model can aid in the desensitization process, a process often necessary in the case of strong allergic reactions or allergies induced by drug therapy. We improved the existing mathematical models on the evolution of the immune system during allergies by offering a more detailed perspective of the biological workings (we introduced time delays and considered that the body is partitioned into two relevant compartments). The stability study performed, along with the numerical simulations and biological interpretations presented, show how one can obtain scenarios for treatment without allergic reactions. The model can be used to find the optimal allergen dose for desensitization and to adjust it accordingly throughout the desensitization process.

Although our focus was on drug-induced allergies, the model is of a versatile nature and can be adapted to describe other types of allergies.

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References

1. Castells, M. Drug hypersensitivity and anaphylaxis in cancer and chronic inflammatory diseases, the role of desensitizations. *Front. Immunol.* **2017**, *8*, 1472. [[CrossRef](#)] [[PubMed](#)]
2. Abdullah, R.; Badralexi, I.; Halanay, A. Stability Analysis in a New Model for Desensitization of Allergic Reactions Induced by Chemotherapy of Chronic Lymphocytic Leukemia. *Mathematics* **2023**, *11*, 3225. [[CrossRef](#)]
3. Berger, A. Th1 and Th2 responses: What are they? *BMJ* **2000**, *321*, 424. [[CrossRef](#)] [[PubMed](#)]
4. Gubernatorova, E.; Gorshkova, E.; Namakanova, O.; Zvartsev, R.; Hidalgo, J.; Drutskaya, M.; Tumanov, A.; Nedospasov, S. Non-redundant functions of IL-6 produced by macrophages and dendritic cells in allergic airway inflammation. *Front. Immunol.* **2018**, *9*, 2718. [[CrossRef](#)] [[PubMed](#)]
5. Brahmakshatriya, V.; Kuang, Y.; Devarajan, P.; Xia, J.; Zhang, W.; Vong, A.M.; Swain, S.L. IL-6 Production by TLR-Activated APC Broadly Enhances Aged Cognate CD4 Helper and B Cell Antibody Responses In Vivo. *J. Immunol.* **2017**, *198*, 2819–2833. [[CrossRef](#)] [[PubMed](#)]
6. Dienz, O.; Rincon, M. The effects of IL-6 on CD4 T cell responses. *Clin. Immunol.* **2009**, *130*, 27–33. [[CrossRef](#)] [[PubMed](#)]
7. Hong, D. Desensitization for allergic reactions to chemotherapy. *Yonsei Med. J.* **2019**, *60*, 119. [[CrossRef](#)]
8. Sancho-Serra, M.D.; Simarro, M.; Castells, M. Rapid ige desensitization is antigen specific and impairs early and late mast cell responses targeting fceri internalization. *Eur. J. Immunol.* **2011**, *41*, 1004–1013. [[CrossRef](#)]
9. Etemadifar, M.; Salari, M.; Saeri, M.; Sigari, A.A.; Ebrahimi, S. Rituximab induced cytokine release syndrome in an MS patient: A case report. *Clin. Case Rep.* **2021**, *9*, e04407. [[CrossRef](#)]
10. Chen, W. Pharmacokinetic and Pharmacodynamic Modeling of Erythropoiesis. Ph.D Thesis, Drexel University, Philadelphia, PA, USA, 2010; pp. 27–30.
11. Segel, L.A.; Fishman, M.A. Modeling immunotherapy for allergy. *Bull. Math. Biol.* **1996**, *58*, 1099–1121.
12. Dambeck, H.; Metzner, G.; Behn, U. Modeling Th1-Th2 regulation, allergy, and hyposensitization. In *Dynamical Modeling in Biotechnology*; Lectures Presented at the EU Advanced Workshop; World Scientific: Singapore, 2001; pp. 227–243.
13. Fouchet, D.; Regoes, R. A Population Dynamics Analysis of the Interaction between Adaptive Regulatory T Cells and Antigen Presenting Cells. *PLoS ONE* **2008**, *3*, e2306. [[CrossRef](#)] [[PubMed](#)]
14. Gross, F.; Metzner, G.; Behn, U. Mathematical modeling of allergy and specific immunotherapy: Th1, Th2 and Treg interactions. *J. Theor. Biol.* **2011**, *269*, 70–78. [[CrossRef](#)] [[PubMed](#)]
15. Vogel, R.; Behn, U. Th1–Th2 Regulation and Allergy: Bifurcation Analysis of the Non-Autonomous System. *Math. Model. Biol. Syst.* **2008**, *2*, 145–155.
16. Diehl, S.; Anguita, J.; Hoffmeyer, A.; Zupton, T.; Ihle, J.N.; Fikrig, E.; Rinco, M. Inhibition of Th1 Differentiation by IL-6 is Mediated by SOCS1. *Immunity* **2000**, *13*, 805–815. [[CrossRef](#)] [[PubMed](#)]
17. Friedman, A.; Liao, K.L.; Bai, X.F. Mathematical modeling of interleukin-27 induction of anti-tumor t cells response. *PLoS ONE* **2014**, *9*, e91844.
18. Wu, G. Calculation of steady-state distribution delay between central and peripheral compartments in two-compartment models with infusion regimen. *Eur. J. Drug Metab. Pharmacokinet.* **2002**, *27*, 259–264. [[CrossRef](#)] [[PubMed](#)]
19. Kharitonov, V.L. *Time-Delay Systems: Lyapunov Functionals and Matrices*; Springer: Berlin/Heidelberg, Germany, 2013.

20. Domoshnitsky, A.; Friedman, E. A positivity-based approach to delay-dependent stability of systems with large time-varying delays. *Syst. Control Lett.* **2016**, *97*, 139–148. [[CrossRef](#)]
21. Rumyantsev, V.V.; Vrotnikov, V.I. *Foundations of Partial Stability and Control*; Publisher: Nijnii Tagil, Russia, 2014. (In Russian)
22. Aleksandrov, A.; Aleksandrova, E.; Zhabko, A.; Chen, Y. Partial stability analysis of some classes of nonlinear systems. *Acta Math. Sci.* **2017**, *37*, 329–341. [[CrossRef](#)]
23. Corduneanu, C. On partial stability for delay systems. *Ann. Pol. Math.* **1975**, *29*, 357–362. [[CrossRef](#)]
24. Halanay, A. *Differential Equation: Stability, Oscillations, Time-Lags*; Academic Press: New York, NY, USA, 1966.
25. Hatvani, L. On partial asymptotic stability and instability, I (Autonomous systems). *Acta Sci. Math.* **1983**, *45*, 219–231.
26. Vrotnikov, V.I. *Partial Stability and Control: The State-of-the-Art and Development Prospects*. Ph.D. Thesis, Ural State Technical University, Nizhni Tagil, Russia, 2003.
27. Kim, P.; Lee, P.; Levy, D. A theory of immunodominance and adaptive regulation. *Bull. Math. Biol.* **2010**, *73*, 1645–1665. [[CrossRef](#)]
28. Nazari, F.; Pearson, A.T.; NoÈ, J.E.; Jackson, T.L. A mathematical model for IL-6-mediated, stem cell driven tumor growth and targeted treatment. *PLoS Comput. Biol.* **2018**, *14*, e1005920. [[CrossRef](#)]
29. Silva, J.; Morais, R.; Rodrigues da Silva, I.; Adimy, M.; Mancera, P.A. Mathematical model for treatment of papillary thyroid cancer using the allee effect. *J. Biol. Syst.* **2020**, *28*, 701–718. [[CrossRef](#)]

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